

of Questions: 30

Question #: 1

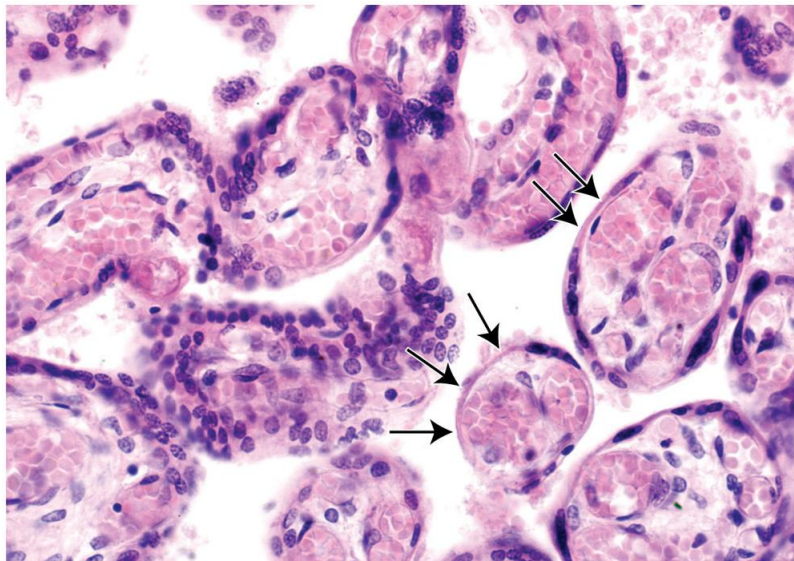
An 18-year-old woman is brought to the emergency department because of a 2-day history of heavy vaginal bleeding. She is at 16 weeks' gestation and has a history of high blood pressure. Physical examination shows blood pooled in the posterior fornix of the vagina and an enlarged uterus. Laboratory studies show a positive serum beta human chorionic gonadotropin test (pregnancy test). An ultrasound shows a thickened endometrium with debris in the uterine cavity. A diagnosis of spontaneous abortion with retained fetal tissue is made. Surgical clearing of the endometrium is carried out to alleviate the bleeding. A photomicrograph of tissue obtained from uterus is shown. Which of the following hormones is secreted by the indicated structure?

- A. Melatonin
- B. Thyroxine
- ✓C. Progesterone
- D. Androstenedione
- E. Follicle stimulating hormone

Rationale: The image shows tertiary chorionic villi, which is part of the placenta. The placenta functions as an endocrine organ, producing steroid and peptide hormones. Immunocytochemical studies indicate that the syncytiotrophoblast is the site of synthesis of these hormones.

The steroid hormones progesterone and estrogen have essential roles in the maintenance of pregnancy. Just about the end of the 8th week, the placenta produces enough progesterone to maintain the pregnancy.

Attachment:



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Item Description: HCB.1167 - Placenta secretion

Question #: 2

A 19-year-old man is brought to the emergency department by the paramedics because of difficulty closing his jaw, for past 2 hours. Physical examination shows dislocated mandible. Radiological examinations shows mandibular head is displaced beyond the articular tubercle of the temporomandibular joint. Which of the following muscles most likely contracted as a result of this condition?

- A. Masseter
- B. Temporalis
- C. Medial pterygoid
- ✓D. Lateral pterygoid

Rationale: The lateral pterygoid muscles act bilaterally to protract the jaw and unilaterally to rotate it. Because the fibers of the lateral pterygoid muscle insert on the anterior aspect of the temporomandibular joint's articular disk as well as the head of the mandible, spasm of this muscle, such as during a yawn, can cause dislocation of the mandible by pulling the disk anterior to the articular tubercle. Reduction is achieved by pushing the mandible backwards and downwards until the mandible's head reenters the mandibular fossa.

Item Description: Lateral pterygoid muscle

Question #: 3

A 21-year-old woman is brought to the emergency department following a car crash where she impacts the steering wheel. Physical examination shows abdominal bruising and guarding on palpation. Imaging studies show a lacerated pancreas with the digestive enzymes spilling anterior to it. Into which of the following locations will the contents of the lacerated organ most likely spill?

- A. Supracolic part of the greater sac
- ✓B. Omental bursa
- C. Subhepatic bursa
- D. Infracolic part of the greater sac
- E. Subhepatic recess

Rationale: The omental bursa or lesser sac is located posterior to the stomach and anterior to the pancreas. Therefore, since the pancreas is lacerated, and the contents are spilling anterior to it, they will spill into the space anterior to the pancreas, which is the omental bursa.

Item Description: A 21-year-old woman

Question #: 4

A 1-month old boy is brought to the physician by his mother for a routine examination. She points out a small opening in the skin just in front of his left ear. The physician tells her not to be concerned because this is simply the result of a harmless condition in the development of the external ear. Which of the following is the most likely diagnosis?

- ✓A. Pre-auricular sinus
- B. Keloid
- C. Auricular hillock
- D. Auricular cyst
- E. Auricular fistula

Rationale: During development of the ear small pits may appear around the area. When anterior they are referred to as pre-auricular sinuses or pits.

Item Description: A 1-month old boy is

Question #: 5

A 22-year-old man is brought to the emergency department because of severe bleeding from a head injury after a car crash. Physical examination shows the three fused layers of the scalp has been peeled back as a single layer. Which of the following layers are most likely involved?

- A. Skin, aponeurosis, pericranium
- B. Skin, loose connective tissue, pericranium
- C. Skin, loose connective tissue, aponeurosis
- ✓D. Skin, dense connective tissue, aponeurosis
- E. Skin, loose connective tissue, dense connective tissue

Rationale: The first three layers of the scalp - skin, dense connective tissue and the aponeurosis can be considered as the scalp proper and are intimately attached. They can therefore be peeled back as a single layer.

Item Description: A 22-year-old man is

Question #: 6

An 18-year-old man is brought to the emergency department after a crash where his forehead absorbed most of the impact. Imaging studies show multiple cranial fractures. Physical examination shows a clear nasal discharge indicating a fracture through the anterior cranial fossa. Which of the following nerves is most likely damaged?

- A. Optic
- B. Facial
- ✓C. Olfactory
- D. Hypoglossal
- E. Maxillary

Rationale: Fracture of the anterior cranial fossa leading to clear discharge from the nose is an indication that the ethmoid bone, in particular the cribriform plate is fractured. This will damage the olfactory nerves passing through the foramina in

the cribiform plate.

Item Description: An 18-year-old man i

Question #: 7

A 15-year-old boy is brought to the physician by his mother because he is unable to rotate his head to the left side. He suffered a stab wound to his neck just posterior to the sternocleidomastoid 3 weeks ago. Which of the following nerves was most likely damaged?

- ✓A. Right spinal accessory
- B. Left spinal accessory
- C. Right vagus
- D. Left vagus
- E. Ansa cervicalis

Rationale: Correct A: The sternocleidomastoid muscle rotates the head to the opposite side. Therefore since the patient is unable to rotate his head to the left, the right sternocleidomastoid muscle is not functioning. This is innervated by the right spinal accessory nerve.

Incorrect :

B: the left spinal accessory innervates the left sternocleidomastoid which is not affected in this patient.

C & D: The vagus nerve does not innervate the sternocleidomastoid muscles.

E: The ansa cervicalis is the motor part of the cervical plexus. It innervates the infrahyoid muscles which includes the omohyoid, sternohyoid and sternothyroid. The thyrohyoid is innervated by C1 via the hypoglossal nerve.

Item Description: Spinal accessory nerve

Question #: 8

A 15-year-old girl comes to the physician because of a 24-hour history of headache, facial pain along her cheekbones and vision problems. She has an infected pimple on the ala of her nose which she has been repeatedly pick at. She is diagnosed with cavernous sinus thrombosis. Which of the following veins is the

most likely route that introduced the infection to the cranial cavity?

- ✓A. Angular vein
- B. Superior sagittal sinus
- C. Lingual vein
- D. Superior petrosal sinus
- E. Internal jugular veins

Rationale: The danger area of the face lies between the deep facial and angular veins. Infection of the upper lip, vestibule of the nose and eyelids spread by way of the angular, supraorbital and supratrochlear veins to the ophthalmic veins which drain directly into the cavernous sinus.

Item Description: Danger triangle of face

Question #: 9

A 25-year-old man comes to the physician because of numbness of the lower lip, left side of his face just anterior to his ear and difficulty chewing his food, for past 5 days. The nerve responsible for innervation to the affected region most likely passes through which of the following foramen?

- ✓A. Foramen ovale
- B. Foramen rotundum
- C. Internal acoustic meatus
- D. Stylomastoid foramen
- E. Mandibular foramen

Rationale: Correct: This is the mandibular division of the trigeminal nerve which enters the infratemporal fossa through the foramen ovale.

Incorrect answers

B is the maxillary division of the trigeminal nerve enters the pterygopalatine fossa through the foramen rotundum.

C and D: The facial nerve exits the cranium via the internal acoustic meatus. The motor branch of the facial nerve exits onto the face via the stylomastoid foramen.

E: the inferior alveolar nerve, which is a branch of the mandibular nerve, as well as

the inferior alveolar artery, enter the mandibular foramen traveling through the mandibular canal and exit at the mental foramen on the anterior mandible at which point the nerve is known as the mental nerve.

Item Description: Trigeminal nerve route

Question #: 10

A 50-year-old man comes to the emergency department because of a 2-week history of a pustular, draining abscess on his lower lip. He says he punctured the middle of his lip with a rusty nail and the wound is not closing spontaneously. Physical examination shows an open abscess. Which of the following lymph nodes is most likely to be enlarged on palpation?

- A. Inferior deep cervical
- B. Occipital
- C. Preauricular
- ✓D. Submental
- E. Superior deep cervical
- F. Supraclavicular

Rationale: Collar nodes (superficial): • occipital • mastoid (posterior auricular) • parotid (pre-auricular) • submandibular • submental afferents drain superficial structures of face, scalp, and head form a collar around the neck and efferents drain mainly to the deep cervical nodes

Superficial cervical nodes - organized into anterior group (along anterior jugular v.) and posterolateral group (along EJV) - afferent drain superficial structures of the neck – efferent drain to the deep cervical nodes

Supraclavicular lymph nodes are lymph nodes found above to the clavicle, that can be felt in the supraclavicular fossa.

Item Description: Neck lymph nodes

Question #: 11

A 50-year-old man comes to the physician because of a 1-week history of painful salivation. Physical examination shows a stone blocking the right submandibular duct and surgery is scheduled to remove the stone. During the surgery, there is trauma to the nerve that wraps around the submandibular duct. Which of the following symptoms will most likely be present in the patient as a result of this nerve damage?

- A. Weakened ability to protrude the tongue
- B. Weakened ability to depress the mandible
- C. Weakened ability to elevate the mandible
- ✓D. Decreased general sensation from the anterior 2/3rds of the tongue on the affected side
- E. Decreased general sensation from the posterior 1/3rd of the tongue on the affected side

Rationale: Correct answer: The relationship between lingual nerve and submandibular duct begin with nerve lateral to the submandibular duct, this nerve courses anteromedially by looping beneath the duct and then terminating as several medial branches. The lingual nerve carries general sensation from the anterior two thirds of the tongue. Fibers for taste sensation travel with the lingual nerve distal to its junction with the chorda tympani branch of the facial nerve. Incorrect answers: general sensation from the posterior 3rd of the tongue is carried via the glossopharyngeal nerve. Protrusion of the tongue is a function of the hypoglossal nerve. Muscles of mastication are responsible for depressing and elevating the mandible and they are supplied by motor branches of the mandibular nerve.

Item Description: Lingual nerve pathway

Question #: 12

A 40-year-old man come to the physician because of a 3-day history of loud sounds, impaired taste sensation and dry mouth. He was struck in the head by flying debris shortly before his current symptoms. Physical examination shows flattening of the nasolabial folds. In which of the following labeled areas (see image) is the lesion most likely located?

- A. A
- ✓B. B
- C. C
- D. D
- E. E

Rationale: Correct answer : B - As the facial nerve travels through the medial and posterior walls of the middle ear, it gives off multiple branches including chorda tympani (taste sensations from anterior two-thirds of the tongue, and parasympathetic innervation to the submandibular and sublingual glands). Other branches are nerve to stapedius muscle (stabilizes the stapes bone to help limit loud noises), and greater petrosal nerve (parasympathetic to lacrimal gland, and the mucous membrane of the nasal cavity and the palate). It then exits through the stylomastoid foramen as the motor branch to the muscles of facial expression. The patient's presentation indicates loss of function of nerve to stapedius, chorda tympani, and the motor branch. A lesion at B, will affect all these branches.

Incorrect answers:

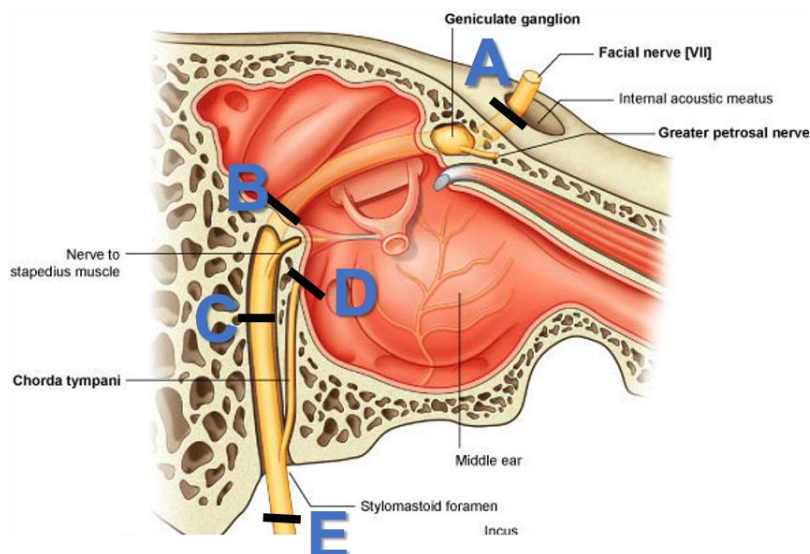
A - Injuries proximal to origin of greater petrosal nerve at the internal acoustic meatus will present with dry eyes and dry nasal cavity, in addition all of the other symptoms.

C - Injuries distal to origin of nerve to stapedius are unlikely to present with hyperacusis.

D - Isolated injuries to chorda tympani are unlikely to present with hyperacusis.

E - Injury at the stylomastoid foramen will lead to loss of the nasolabial fold. but is unlikely to present with hyperacusis and decreased taste sensation to the anterior 2/3 of the tongue.

Attachment:



Item Description: Facial Nerve pathway through middle ear

Question #: 13

A 16-year-old boy is brought to the physician by his parents because of a 1-day history of fever, drowsiness and a confused mental state. He is admitted to hospital and diagnosed with a cavernous sinus infection and thrombosis resulting from an abscess in the infratemporal fossa. Which of the following is most likely the path taken by the infection from the fossa to the cavernous sinus?

- A. Ophthalmic vein
- B. Facial vein
- ✓C. Pterygoid venous plexus
- D. Occipital emissary vein
- E. Maxillary vein

Rationale: The pterygoid venous plexus is located in the infratemporal fossa and connect to the cavernous sinus via emissary veins. Therefore infections can spread from the infratemporal fossa to the cavernous sinus.

Item Description: A 16-year-old boy is

Question #: 14

A 44-year-old man comes to the physician because of a 1-week history of drooping of the right corner of his mouth. He can clench his teeth and chew on demand. Physical examination shows slight hearing loss and difficulty maintaining his balance while standing on one foot. His blink reflex is absent on the right, but sensation is normal on his entire face. Imaging studies show a tumor in the posterior cranial fossa. Which of the following additional symptoms is most likely present?

- A. Loss of touch sensation on the anterior two thirds of the tongue
- B. Mydriasis
- C. Loss of taste on the posterior third of the tongue

- D. Inability to shrug the shoulders
- ✓E. Decreased ability to produce tears

Rationale: Based on the symptoms described both the facial and vestibulocochlear nerves are affected. This indicates that there is a tumor located by the internal acoustic meatus. The only other symptom that would occur as a result is the inability to produce tears.

Item Description: A 44-year-old man co

Question #: 15

A 22-year-old woman is brought to the emergency department because of a head wound after she fainted. Physical examination shows a 4cm deep laceration to the back of her head that is bleeding freely. Branches of which of the following arteries is most likely damaged?

- ✓A. Occipital
- B. Deep temporal
- C. Posterior auricular
- D. Vertebral
- E. Superficial temporal

Rationale: The occipital artery is a branch of the external carotid artery, which supplies blood to the posterior scalp.

Item Description: A 22-year-old woman

Question #: 16

A 65-year-old man come to the physician because of a small lump on the right side of face, anterior to his right ear. At first the physician thought it might be just an enlarged lymph node. Physical examination shows no sign of infection around ear and scalp area, but the small lump is superficial to the parotid salivary gland, anterior to the external ear and posterior to the masseter muscle. Diagnosis of parotid gland cancer is made. Which of the following is the most likely risk of surgical removal of tumor?

- A. Lose of taste on the right side of tongue
- B. Lose of chewing on the right side
- C. Tongue deviating to the right side on protrusion
- ✓D. Right side facial paralysis

Rationale: The parotid salivary gland is where approximately 80% of salivary gland tumors begin. The stylomastoid foramen is where the facial nerve exits. This nerve then splits into 5 to 7 major branches that control facial expression muscles. While passing through the substance of the parotid salivary gland, the branches split into temporal, zygomatic, buccal, mandibular, and cervical. While all of these branches are unlikely to be cut during the removal of the parotid tumor, they are all at risk.

Item Description: Parotid gland cancer

Question #: 17

A 4-year-old girl is brought to the physician by her parents because of a 3-month history of easy fatigability while playing. Her parents say that she has a habit of sucking on yellow paint chips she finds in their old barn. Physical examination shows conjunctival pallor. Which of the following enzymes is most likely inhibited in this child?

- ✓A. Delta-aminolevulinic acid dehydratase
- B. Xanthine oxidase
- C. Glutathione peroxidase
- D. Delta-aminolevulinic acid synthase
- E. Dopamine beta-hydroxylase
- F. Homocysteine methyl transferase

Rationale: The vignette describes a child with lead poisoning (paint). Lead inhibits Heme synthesis by inhibiting delta-ALA dehydratase (porphobilinogen synthase). Ferrochelatase in heme biosynthesis is also inhibited by lead.

Item Description: Heme biosynthesis

Question #: 18

A 20-year-old man comes to the physician because of a 1-month history of seizures. He is prescribed valproic acid which acts by inhibiting the GABA-transaminase enzyme and increasing levels of GABA. Which of the following enzymes is most likely required in the presynaptic neurons for the synthesis of this neurotransmitter?

- A. Monoamine oxidase
- B. DOPA decarboxylase
- ✓C. Glutamate decarboxylase
- D. Acetylcholinesterase
- E. Histidine decarboxylase
- F. Tryptophan hydroxylase

Rationale: GABA-Transaminase (GABA-T) is involved in the degradation of GABA to succinate which enters the Krebs cycle to form alpha-ketoglutarate which is converted to glutamate (via glutamate dehydrogenase) and then glutamine (by glutamine synthetase). The inactivation of GABA takes place in the glial cells. Glutamine is next transported to GABA-ergic neurons, where it is converted to glutamate (via glutaminase) which then forms GABA (via glutamate decarboxylase; PLP requiring enzyme).

So, administration of valproic acid would increase GABA levels by inhibiting its degradation.

Monoamine oxidase (MAO) inhibitors inhibit degradation of serotonin and catecholamines.

Acetylcholinesterase inhibitors inhibit degradation of acetyl choline.

Histamine is formed from histidine by histidine decarboxylase (PLP requiring enzyme).

DOPA decarboxylase converts DOPA to dopamine which is then used for the formation of norepinephrine and epinephrine.

Tryptophan hydroxylase (requires tetrahydrobiopterin) and is required for the formation of Serotonin and melatonin.

Item Description: Glutamate decarboxylase

Question #: 19

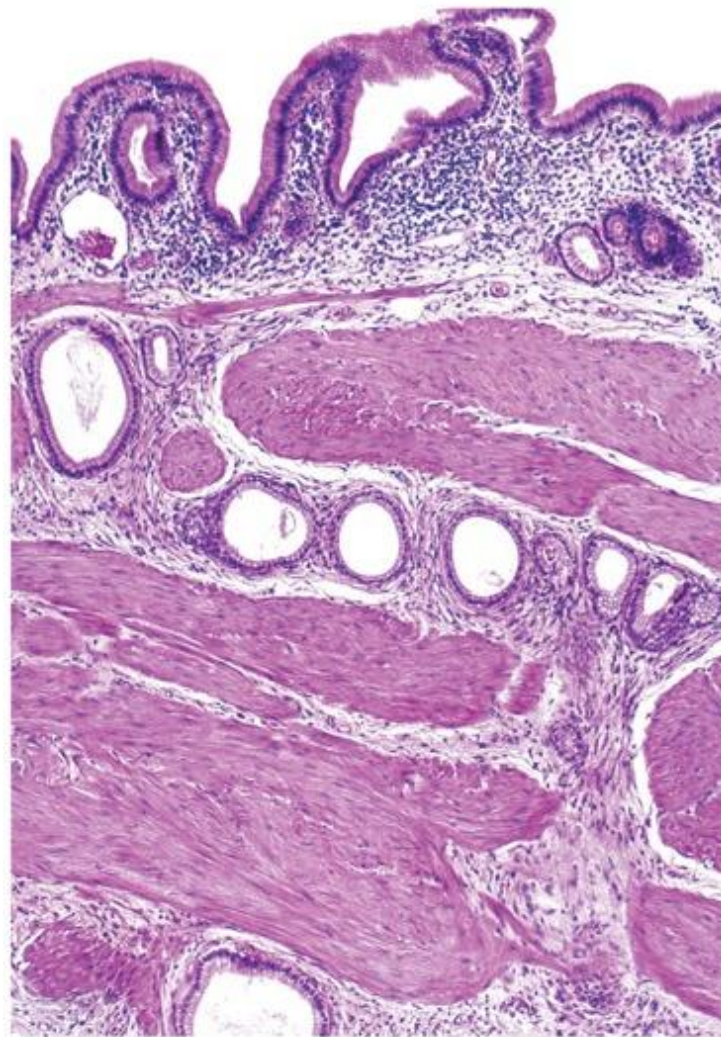
A 45-year-old woman comes to the physician because of a 2-month history of intermittent upper abdominal discomfort and bloating which occurs after meals. Physical examination of the abdomen shows mild tenderness in the right upper quadrant. Imaging studies show enlargement and thickening of the wall of a

particular gastrointestinal organ. The organ is removed via laparoscopic surgery. A photomicrograph of the resected organ is shown. Which of the following gastrointestinal structures is most likely affected?

- A. Stomach
- B. Duodenum
- C. Jejunum
- ✓D. Gallbladder
- E. Ileum
- F. Esophagus

Rationale: This light micrograph is of the gallbladder. Notice the extensively folded mucosa lined by simple columnar epithelium, and the irregular arrangement of the muscle tissue.

Attachment:



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Item Description: HCB.1296 - Structure of the gall bladder

Question #: 20

A 24-year-old man comes to the physician because of a 3-month history of intermittent headaches, weight gain, and change in his shoe size. His blood pressure is 146/94 mmHg; other vital signs are within normal limits. Physical examination shows coarse skin, prominent nose, protruding lower jaw, and enlarged hands and feet. MRI of the brain shows an 8-mm mass in the Sella turcica. Laboratory studies show increased plasma IGF-1. Which of the following is most likely decreased in this patient?

- A. Plasma growth hormone
- B. Plasma free fatty acids
- C. Synthesis of proteins in skeletal muscles
- ✓D. Sensitivity to an oral glucose load
- E. Uptake of amino acids by skeletal muscle

Rationale: This patient has a tumor that secretes excess growth hormone. Excess growth hormone increases IGF-1, amino acid uptake & protein synthesis, FFA levels and blood glucose. In a normal person, hypoglycemia stimulates GH secretion and an oral glucose load (hyperglycemia) inhibits (decreases) GH secretion. However, the same oral load of glucose will not decrease plasma GH in a patient with a tumor that secretes excess GH. Administration of a glucose load does not suppress plasma GH in this case because the sensitivity of the gland to oral glucose is decreased.

Item Description: PHYS.1013. Glucose load in acromegaly

Question #: 21

An 80-year-old woman comes to the physician because of a 3-month history of fatigue and shortness of breath. Her vital signs are within normal limits. Physical examination shows a pale tongue and conjunctiva. Laboratory studies show low hemoglobin concentration, elevated mean corpuscular volume, and a low reticulocyte count. A peripheral blood smear shows macrocytic hypochromic red cells and hypersegmented neutrophils. Gastric juice studies show a decreased basal and maximal acid secretion. She is diagnosed with megaloblastic anemia

due to intrinsic factor deficiency. Which of the following cells in the stomach is most likely defective in this patient?

- A. Chief cells
- B. G cells
- C. Goblet cells
- D. Mucous neck cells
- ✓E. Parietal cells

Rationale: The patient presents with symptoms indicative of an anemia due to IF deficiency. IF is secreted by the parietal cells that also secrete acid in the stomach.

Item Description: PHYS.1123. Intrinsic factor cellular source

Question #: 22

A 48-year-old woman is brought to the emergency department because of a 30-minute history of a deep penetrating wound to her right knee. Prior to surgery, a local anesthetic is administered via lumbar puncture, leading to analgesia and paralysis. Which of the following events is most likely directly affected by the drug?

- A. Resting membrane potential
- B. After-hyperpolarization
- C. Receptor potential
- D. Postsynaptic potential
- ✓E. Action potential

Rationale: Lidocaine blocks voltage-gated sodium channels, thereby preventing action potentials.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0035 Action potential lidocaine

Question #: 23

A 45-year-old woman comes to the physician because of a 6-hour history of a traumatic injury to the back of her head. Over 24 hours, confusion and weakness arise. CT scan of the head shows a crescent-shaped hematoma that crosses

suture lines, compressing the brain parenchyma. Between which of the following structures is the blood most likely accumulating in this patient?

- ✓A. Dura mater and arachnoid
- B. Arachnoid and pia
- C. Inner and outer dura layers
- D. Pia mater and brain parenchyma
- E. Skull and dura mater

Rationale: The crescent-shaped collection that crosses suture lines, particularly with a progressive worsening of symptoms after trauma, is characteristic of a subdural hematoma. Subdural hematomas occupy the potential space between the dura mater and the arachnoid, typically due to the tearing of bridging veins.

Item Description: SOM.MK.I.BPM2.2.NB.1.NSCI.0071 Intracranial hemorrhages

Question #: 24

A 46-year-old man comes to the physician because of a 2-day history of severe pain in his right knee. He says that his knee briefly "locked" when he got out of his car. He undergoes arthroscopic surgery for repair, requiring the application of a local anesthetic via lumbar puncture resulting in anesthesia and paralysis of his lower body. Which of the following structures is most likely targeted by the anesthetic?

- A. Nicotinic receptors
- B. Muscarinic receptors
- ✓C. Voltage-gated sodium channels
- D. Ligand-gated calcium channels
- E. Voltage-gated calcium-channels
- F. Pressure-gated sodium channels
- G. Ligand-gated chloride channels
- H. Glycine receptors

Rationale: Local anesthetics such as lidocaine interact with voltage-gated sodium channels to suppress the expression of action potentials. when administered peripherally, the drugs can produce both anesthetic and paralytic effects. Note that autonomic targets can also be chemically denervated by the local anesthetics, leading to urinary retention, for example.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0035 Explain the action-potential blocking effects of lidocaine.

Question #: 25

A group of researchers investigates relations among ionic conductance and membrane potential. A drug that blocks potassium channels is applied to a cultured neuron. Which of the following outcomes is most likely?

- ✓A. Depolarization
- B. Hyperpolarization
- C. Increased conduction velocity
- D. Decreased conduction velocity
- E. Initiation of action potentials
- F. Prevention of action potentials

Rationale: Reduced efflux of potassium should drive the membrane potential away from the equilibrium potential of potassium. The closing of the K⁺ channels should thus exert a depolarizing influence. It is unclear whether the depolarization would be sufficient to trigger action potentials.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0027 Excitable cells signal conversion

Question #: 26

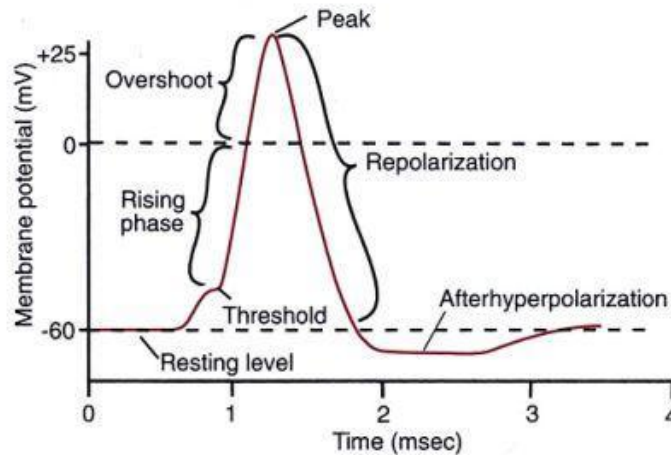
A researcher is studying the excitability of basal ganglia neurons. He finds that during a certain portion of the action potential it is possible to stimulate the neuron and produce another action potential but only if a greater-than-normal excitatory input is applied. Which of the following events most likely overlaps with the indicated period of reduced excitability (see figure)?

- A. Rest
- ✓B. After-hyperpolarization
- C. Overshoot
- D. Repolarization
- E. Rising phase

F. Peak

Rationale: The component of one action potential during which initiation of another action potential requires greater-than-normal excitatory synaptic input is called the relative refractory period. During this period, voltage-gated sodium channels have reset, but potassium currents still predominate. This period overlaps with the last portions of the repolarization and the after-hyperpolarization phase.

Attachment:



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Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0038 Action potential conductances

Question #: 27

A 21-year-old woman comes to the physician because of a 6-hour history of painful cutaneous burns to her back caused by ultraviolet radiation. The sensation is of a throbbing quality. In such cases, propagation of the pain-eliciting signals from the skin to the spinal cord is slow but continuous. Which of the following pairs of properties most likely applies to the cells involved in the transmission of injury-related action potentials to the spinal cord?

- A. Pseudounipolar; myelinated
- ✓B. Pseudounipolar; unmyelinated
- C. Multipolar; myelinated
- D. Multipolar; unmyelinated
- E. Bipolar, myelinated

F. Bipolar, unmyelinated

Rationale: The most slowly conducting axons are unmyelinated and of small diameter. Were the pain sharp with stabbing qualities, then it is much more likely that more rapidly conducting small myelinated axons would be involved. As somatosensory fibers are under consideration, the cellular morphotype is pseudounipolar.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0034 Conduction velocity myelination

Question #: 28

A 61-year-old man is brought to the emergency department by his son because of a 3-hour history of tingling, numbness and loss of control of his legs. Laboratory studies and nerve conduction test results are shown in the tables below. Which of the following is the most likely diagnosis?

Parameter	Measured value	Normal Range
K+	4.1 mM/L	3.5-5.0
Na+	143 mM/L	136-145
Cl-	98 mM/L	95-105

Test	Measured Value	Normal value
Peripheral nerve conduction velocity	20 m/s	40-60

- A. Hyperkalemia
- B. Hypernatremia
- ✓C. Demyelination disorder
- D. Transient ischemic attack
- E. Traumatic injury

Rationale: All the ion concentrations are within normal ranges, so there is no indication of any ion balance disturbance, which rules out 3 of the answers. A transient ischemic attack (TIA) would be more likely to present with focal and temporary paralysis and does not explain the slowed nerve conduction speed (should be 40-60 m/s). The slowed peripheral nerve conduction makes a demyelination disorder most likely, especially peripherally (e.g., Guillain-Barré

Syndrome).

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0034 Conduction velocity myelination

Question #: 29

A 63-year-old man comes to the physician because of a 1-year history of fatigue and pain. He says that over-the-counter preparations are ineffective in controlling pain. He receives a prescription for a drug that inhibits the uptake of serotonin. Which of the following nuclei is most likely integral to the synthesis of the targeted transmitter?

- A. Locus ceruleus
- B. Substantia nigra
- C. Nucleus basalis of Meynert
- ✓D. Nucleus raphe magnus
- E. Nucleus accumbens

Rationale: The nucleus raphe magnus mediates the release of serotonin in the dorsal horn of the spinal cord and also the spinal nucleus of the trigeminal nerve. In this way, serotonin mitigates synaptic transmission normally arising from pain fibers entering the spinal cord or brainstem. The other nuclei listed are not intimately involved in serotonergic transmission.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0107 Catecholamines and serotonin

Question #: 30

A 23-year-old woman comes to the physician because of a 5-year history of poorly controlled seizures. She is prescribed a new drug that acts as an agonist of an inhibitory amino acid. Which of the following neurotransmitters is most likely targeted by this new drug?

- A. Serotonin
- B. Dopamine

- ✓C. GABA
- D. Glutamate
- E. Norepinephrine

Rationale: Two of the transmitters listed are amino acid transmitters. Glutamate tends to exert excitatory effects, whereas GABA tends to exert inhibitory effects when interacting with its receptors.

Item Description: SOM.MK.I.BPM2.1.NB.1.NSCI.0106 Amino acid transmitters